

THE ABERDARE TECHNICAL COLLEGE
ICT TECHNICIAN LEVEL 5

UNIT OF COMPETENCE: NETWORK DESIGN AND MANAGEMENT

UNIT CODE: IT/CU/ICTA/CR/02/5/MA

CONTINUOUS ASSESSMENT TEST (CAT) 1, TERM 3

TVET CDACC ICT LEVEL 5

Time: 2 Hours
Marks: 50 Marks

NAME Ben Mann DATE 25/5/26

CDACC REGNO.....

INSTRUCTIONS

- ✓ Answer all questions.
- ✓ Write your answers clearly.
- ✓ Show all workings where necessary.
- ✓ Write your students information on the slots provided
- ✓ The test consists THREE pages and NINE questions.
- ✓ DO NOT remove any page from this test

1. Define the following terms: (4 Marks)

a) Half-duplex

b) Node

c) Bandwidth

d) Topology

2. Differentiate between LAN and WAN. (4 Marks)

3. State FOUR advantages of star topology. (4 Marks)

4. Explain the following network topologies: (10 Marks)

a) Star topology

b) Ring topology

c) partial mesh topology

d) Hybrid topology

Define

- 1] Half duplex → Communication mode in which data can travel in both direction, but not at the same time. ✓
- 2] Node → Any device connected to a network. ✓
- 3] Bandwidth → Maximum amount of data that can be transmitted over a network at given time. ✓
- 4] Topology → physical or logical layout of devices in network how nodes are arranged and connected to each other. ✓

- 2] Differentiate between LAN & WAN
- LAN → covers a small geographical area like home, office, or building. ✓
- WAN → spans a large physical distance such as country, continent or entire globe. ✓

- 3] Advantages of star topology.

Good performance ✓

More secure & reliable ✓ compared to what

Easy to install and manage ✓

Easy fault detection ✓

- 4] Explain the following topologies.

- a] Star topology → Every node is connected to a central device such as a hub or switch. ✓
- b] Ring topology → Each device is connected to two others, forming a continuous circular path for signals. ✓
- c] Partial mesh topology → Each device is connected directly to every other device in network. ✓
- d] Hybrid topology → Combination of two or more topologies. ✓
- e] Point to point topology → Consist of a permanent link between two endpoints. ✓

Q1 Components of a Computer network.

- Hub ✓
- Switches ✓
- Router ✓
- NIC ✓

OF

Q2 Functions of a router

- Connects network ✓
- Shares internet connections ✓
- Assigns IP address ✓
- Routes data packets. ✓

OF

Q3 Differentiate between PAN, MAN & WAN

PAN → the smallest network type, typically used for communication among devices belonging to an individual.

MAN → connects users with computer resources in a geographical area larger than a LAN but smaller than a WAN.

Q4 The most suitable network

a) LAN is a local area network is within one building or small area.

- b) Switch ✓
- Router ✓

Network interface card.

